

the network is a directed graph with nodes and edges.



Figure 1: A simple neural network architecture. The input layer has three nodes, the hidden layer has three nodes, and the output layer has three nodes. The network is a directed graph with nodes and edges.

2.1.1. The network is a directed graph.

The network is a directed graph with nodes and edges. The nodes are arranged in layers, and the edges represent the connections between the nodes in adjacent layers.

The input layer has three nodes, the hidden layer has three nodes, and the output layer has three nodes. The network is a directed graph with nodes and edges.

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- 1. The input layer has three nodes, the hidden layer has three nodes, and the output layer has three nodes.
- 2. The network is a directed graph with nodes and edges.
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Define a bilinear form B on V by $B(x, y) = \text{tr}(T(x)y)$ for all $x, y \in V$.

$$B(x, y) = \text{tr}(T(x)y) \quad \text{for all } x, y \in V.$$

Let $\mathcal{B}$ be a basis of V . Define a matrix M by $M_{ij} = B(b_i, b_j)$.

Exercise 1a. Show that M is symmetric.

$$M_{ij} = B(b_i, b_j) = \text{tr}(T(b_i)b_j) = \text{tr}(T(b_j)b_i) = B(b_j, b_i) = M_{ji}.$$

Given a vector $v \in V$, define a linear map $T_v: V \rightarrow V$ by $T_v(x) = vx$ for all $x \in V$. Show that $B(x, y) = \text{tr}(T_v(x)y)$ for all $x, y \in V$.

1b. A possible definition of a bilinear form

Let V be a vector space over a field F . A bilinear form B on V is a map $B: V \times V \rightarrow F$ such that $B(x, y) = B(y, x)$ for all $x, y \in V$ and $B(ax + by, z) = aB(x, z) + bB(y, z)$ for all $x, y, z \in V$ and $a, b \in F$.

$$B(ax + by, z) = aB(x, z) + bB(y, z)$$

Let $\mathcal{B}$ be a basis of V . Define a matrix M by $M_{ij} = B(b_i, b_j)$.

$$M_{ij} = B(b_i, b_j) = \text{tr}(T(b_i)b_j) = \text{tr}(T(b_j)b_i) = B(b_j, b_i) = M_{ji}.$$

Suppose that B is a bilinear form on V . Define a linear map $T_B: V \rightarrow V$ by $T_B(x) = \sum_{i=1}^n B(x, b_i)b_i$ for all $x \in V$.

1c. Bilinear forms and matrices

Let B be a bilinear form on V . Define a matrix M by $M_{ij} = B(b_i, b_j)$.

$$M = \begin{pmatrix} B(b_1, b_1) & B(b_1, b_2) & \dots & B(b_1, b_n) \\ B(b_2, b_1) & B(b_2, b_2) & \dots & B(b_2, b_n) \\ \vdots & \vdots & \ddots & \vdots \\ B(b_n, b_1) & B(b_n, b_2) & \dots & B(b_n, b_n) \end{pmatrix}$$

Suppose that B is a bilinear form on V . Define a linear map $T_B: V \rightarrow V$ by $T_B(x) = \sum_{i=1}^n B(x, b_i)b_i$ for all $x \in V$. Show that M is the matrix of T_B with respect to the basis $\mathcal{B}$.

$$M = \begin{pmatrix} B(b_1, b_1) & B(b_1, b_2) & \dots & B(b_1, b_n) \\ B(b_2, b_1) & B(b_2, b_2) & \dots & B(b_2, b_n) \\ \vdots & \vdots & \ddots & \vdots \\ B(b_n, b_1) & B(b_n, b_2) & \dots & B(b_n, b_n) \end{pmatrix}$$